



Heart Disease Analysis

Phase 3: Project Design Phase

Proposed Solution

Naga Nandini Padidham | GitHub: nandinipadidham-igtm/Heart-Disease-Analysis

Solution Summary

The proposed solution is an end-to-end data analytics pipeline that stores heart disease patient data in MySQL, analyzes it using SQL, visualizes it through an interactive Tableau dashboard and story, publishes it to Tableau Public, and embeds it inside a simple Flask web application for easy access.

Core Components

1. Data Layer

- A 4,500-record heart disease dataset (18 columns) is loaded into a MySQL database (heart_disease_db, table HD).

2. Analysis Layer

- SQL queries compute prevalence rate, demographic breakdowns (age, gender, race), and lifestyle/medical risk correlations (smoking, alcohol, diabetes, stroke).

3. Visualization Layer

- Tableau Desktop turns the analyzed data into KPI cards, charts, and an interactive dashboard with filters.
- A six-part Tableau Story presents a guided narrative: overview → age analysis → smoking/alcohol impact → sleep/lifestyle → medical conditions → key insights & recommendations.

4. Publishing Layer

- The dashboard and story are published to Tableau Public, generating a shareable public link.

5. Web Integration Layer

- A lightweight Flask application embeds the published Tableau view via an iframe, wrapped in a simple branded UI.

Key Features

- Real-time-feeling interactivity through Tableau filters (Gender, Heart Disease, Smoking, Alcohol Drinking).
- Clear KPI summary: Total Patients, Heart Disease Cases, Heart Disease Rate, Average BMI.
- Story-driven insight delivery, moving from data to recommendation.
- Public, shareable access with no login required to view the dashboard.
- Simple, extensible Flask front end that can later host additional features (e.g. prediction models).

Expected Outcome

Stakeholders can, within minutes, identify that heart disease risk rises sharply after age 50, is elevated among diabetics and smokers, and use this to justify targeted screening and prevention programs — all through a self-serve, shareable dashboard rather than a static report.